

M003 Extension — English Worksheet (Intermediate)

Topic: Design Your Own 3D Build (x, y, z) · **Course:** 3D Coordinates · **Level:** Extension (Intermediate, after Week 3) · **Time:** about 45 minutes

Pixel art was flat — two numbers, one wall. This week your build is **solid**. Three numbers. You draw it from the **front** and from **above**, then read both drawings back into the world.

Build your canvas. Run this to lay a blank **15 × 15** floor, then put a **red block at your feet** as your **home spot**:

```
def on_run():
    blocks.fill(WHITE_CONCRETE,
                pos(1, 0, 1),
                pos(15, 0, 15),
                FillOperation.REPLACE)
    player.on_chat("run", on_run)
```

Your build stands on that floor. **x** goes across, **y** goes up, **z** goes away from you — a **15 × 15 × 15** space.

● **Stand on the red block every time you run.** If you move, the build moves too.

1 · Predict 🧠

Read each set of steps. Write a sentence about what you will see.

```
place blue block at (3, 9, 4)
place blue block at (4, 9, 4)
place blue block at (5, 9, 4)
```

Three blocks share $y = 9$ and $z = 4$. What shape do they make, and which number changes?


```
place blue block at (7, 2, 4)
place blue block at (7, 3, 4)
place blue block at (7, 4, 4)
```

These three share $x = 7$ and $z = 4$. What shape do they make, and which number changes?


```
place blue block at (7, 5, 1)
place blue block at (7, 5, 2)
place blue block at (7, 5, 3)
```

These three share $x = 7$ and $y = 5$. Which way does this line run, and which number changes?


```
place green block at (6, 6, 6)
place blue block at (6, 6, 6)
```

Two blocks aim for the same (x, y, z) . What do you see — green, blue, or both? Why?

2 · Spot the Bug 🐛

Each one is broken. Fix it, then explain why the original was wrong.

Bug A — The table's two legs should sit at (2, 1, 2) and (2, 1, 4). Both land in the **same spot**.

```
place brown block at (2, 1, 2)
place brown block at (2, 1, 2)
```

Write the fixed line:

Why was it wrong?

Bug B — This should be a line running **away** from you at $x = 8$, $y = 5$.

```
place black block at (8, 5, 3)
place black block at (9, 5, 3)
place black block at (10, 5, 3)
```

Write the fixed lines:

Why was it wrong? Why does your fix work?

3 · Show Your Work 📷 🎥

Stand on your **home spot** (the red block).

Your five rules.

#	Your design must...
1	fit inside the 15 × 15 × 15 space
2	use 3 or more different block colours
3	use 40 or more blocks
4	be recognisable — someone else can name it without being told
5	be drawn on both grids below before you place a single block

Two drawings, one build. The front view gives you **x** and **y**. The top view gives you **x** and **z**. Together they place every block.

Here is one I made — a tree:

🌳 **Front view — x across, y up**

	1	2	3	4	5
5		🌿	🌿	🌿	
4		🌿	🌿	🌿	
3		🌿	🌿	🌿	
2			🌳		
1			🌳		

🌳 **Top view — x across, z away from you**

	1	2	3	4	5
5					
4		🌿	🌿	🌿	
3		🌿	🌳	🌿	
2		🌿	🌿	🌿	
1					

Read it back. The trunk is at $x = 3$, $z = 3$, and stands from $y = 1$ to $y = 2$. The leaves fill $x = 2$ to 4 and $z = 2$ to 4 , at $y = 3$ to $y = 5$.

Now draw yours. Colour the squares.

Count as you go. How many blocks did you use? How many colours?

Then build it. Read the front view for **x** and **y**. Read the top view for **x** and **z**. Place each block at its (x, y, z).

Check your work. Walk around your build. Look at it from the front, then from above. Does each view match your grid? Write down any block you had to move.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 • Your grids. Hold this paper up to the camera.

2 • Your build. Point the camera. Hold your paper next to the build so both are in the shot.

Fill the blanks:

Today I built ____.

I built it using this code: ____.

In this code I used ____.

The hardest part was ____.

That part was hard because ____.

The most fun part was ____.

Something new I learned was ____.

Write your lines here, then say them in your video.

Submit 

Send this worksheet + your video to teacher on KakaoTalk.